

## LAMPASAS, TX, USA

WMO: 720647

Lat: 31.106N

Lon: 98.196W

Elev: 1214

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00229

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 23.0       | 28.2 | 9.0                           | 9.1 | 36.8 | 13.3 | 11.3 | 37.4 | 19.9                  | 52.3 | 18.3 | 52.7 | 6.3                   | 0    | 0.412 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 22.3                   | 100.2           | 71.5 | 98.9 | 71.6 | 96.8 | 73.0 | 78.5                | 91.0 | 77.5 | 90.1 | 76.6 | 88.3 | 8.3                  | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.4                            | 139.3 | 82.7 | 74.5 | 135.0 | 81.5 | 73.7 | 131.3 | 80.4 | 43.0          | 91.4 | 41.7 | 90.1 | 40.8 | 88.5 | 98.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |     |                    | n-Year Return Period Values of Extreme Temperature |           |     |            |     |            |     |            |     |
|-------------------|------|------|----------------------------|------|-----|--------------------|----------------------------------------------------|-----------|-----|------------|-----|------------|-----|------------|-----|
| 1%                |      | 2.5% | 5%                         | Mean |     | Standard Deviation |                                                    | n=5 years |     | n=10 years |     | n=20 years |     | n=50 years |     |
| Min               |      | Max  |                            | Min  | Max |                    | Min                                                | Max       | Min | Max        | Min | Max        | Min | Max        |     |
| (a)               | (b)  | (c)  |                            | (e)  | (f) | (g)                | (h)                                                | (i)       | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |
| 17.6              | 16.0 | 14.0 |                            | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | DB                         | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | WB                         | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 67.3        | 48.1  | 52.4  | 59.6  | 67.1  | 73.5  | 82.0  | 84.7  | 85.1  | 78.4  | 68.1  | 57.6  | 49.7  | (6)  |
|                                                                        | DBStd       | 15.21       | 9.52  | 11.07 | 9.08  | 7.42  | 6.51  | 3.86  | 3.44  | 3.79  | 5.72  | 8.47  | 9.64  | 9.79  | (7)  |
|                                                                        | HDD50       | 439         | 147   | 95    | 26    | 1     | 0     | 0     | 0     | 0     | 0     | 4     | 39    | 127   | (8)  |
|                                                                        | HDD65       | 1982        | 527   | 366   | 210   | 62    | 14    | 0     | 0     | 0     | 1     | 66    | 253   | 483   | (9)  |
|                                                                        | CDD50       | 6746        | 88    | 163   | 324   | 515   | 730   | 960   | 1075  | 1087  | 853   | 566   | 268   | 117   | (10) |
|                                                                        | CDD65       | 2810        | 2     | 13    | 43    | 126   | 278   | 510   | 610   | 622   | 404   | 162   | 32    | 8     | (11) |
|                                                                        | CDH74       | 32044       | 43    | 115   | 366   | 1138  | 2526  | 5901  | 7926  | 8143  | 4110  | 1484  | 230   | 62    | (12) |
|                                                                        | CDH80       | 15906       | 3     | 17    | 59    | 326   | 910   | 2987  | 4483  | 4673  | 1951  | 459   | 36    | 2     | (13) |
| (14) Wind                                                              | WSAvg       | 6.1         | 5.8   | 6.6   | 7.0   | 7.7   | 7.3   | 6.6   | 5.9   | 5.4   | 5.0   | 5.4   | 5.5   | 5.4   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 31.8        | 1.8   | 2.3   | 2.3   | 2.6   | 4.1   | 3.4   | 2.2   | 2.6   | 2.9   | 3.5   | 2.3   | 2.2   | (15) |
|                                                                        | PrecMax     | 42.5        | 8.0   | 7.8   | 6.8   | 8.9   | 9.2   | 8.7   | 14.7  | 12.9  | 9.3   | 10.1  | 8.0   | 9.9   | (16) |
|                                                                        | PrecMin     | 16.6        | 0.0   | 0.1   | 0.1   | 0.0   | 0.4   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 0.0   | 0.0   | (17) |
|                                                                        | PrecStd     | 6.9         | 1.5   | 1.7   | 1.7   | 2.0   | 2.2   | 2.2   | 2.6   | 2.6   | 2.0   | 2.5   | 1.7   | 1.8   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 78.4  | 81.9  | 85.1  | 92.1  | 95.2  | 101.0 | 105.2 | 103.5 | 99.6  | 92.4  | 84.1  | 79.5  | (19) |
|                                                                        |             | MCWB        | 56.5  | 58.5  | 61.8  | 66.6  | 72.4  | 71.5  | 72.1  | 71.3  | 66.9  | 70.4  | 69.0  | 62.2  | (20) |
|                                                                        | 2%          | DB          | 74.1  | 77.0  | 81.3  | 88.2  | 91.2  | 98.4  | 100.1 | 100.3 | 96.3  | 89.5  | 80.5  | 75.8  | (21) |
|                                                                        |             | MCWB        | 58.6  | 61.5  | 63.0  | 68.2  | 72.0  | 71.5  | 71.8  | 71.9  | 71.3  | 70.6  | 64.6  | 62.9  | (22) |
|                                                                        | 5%          | DB          | 69.9  | 74.2  | 78.0  | 83.3  | 89.2  | 95.5  | 98.4  | 99.1  | 93.5  | 84.4  | 76.2  | 71.0  | (23) |
|                                                                        |             | MCWB        | 57.9  | 61.5  | 63.2  | 66.9  | 71.2  | 73.1  | 72.6  | 71.8  | 72.3  | 69.3  | 64.5  | 61.6  | (24) |
|                                                                        | 10%         | DB          | 63.5  | 70.4  | 75.2  | 80.9  | 84.9  | 92.9  | 96.3  | 97.1  | 90.8  | 81.6  | 72.4  | 66.1  | (25) |
|                                                                        |             | MCWB        | 52.7  | 60.8  | 61.8  | 65.4  | 70.4  | 73.9  | 73.8  | 73.0  | 72.3  | 66.0  | 63.4  | 58.8  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 66.4  | 69.2  | 69.7  | 74.8  | 78.2  | 79.9  | 81.1  | 79.0  | 78.5  | 76.6  | 73.1  | 70.0  | (27) |
|                                                                        |             | MCDB        | 71.9  | 75.3  | 75.8  | 83.2  | 87.3  | 91.7  | 96.0  | 91.0  | 90.6  | 86.0  | 79.8  | 74.3  | (28) |
|                                                                        | 2%          | WB          | 62.2  | 66.9  | 68.1  | 72.5  | 76.2  | 78.1  | 78.4  | 77.5  | 77.0  | 74.6  | 70.1  | 67.0  | (29) |
|                                                                        |             | MCDB        | 70.1  | 71.7  | 75.3  | 79.9  | 83.8  | 90.1  | 92.0  | 91.1  | 88.5  | 81.7  | 74.9  | 72.3  | (30) |
|                                                                        | 5%          | WB          | 58.9  | 64.9  | 66.7  | 70.9  | 74.7  | 77.0  | 77.3  | 76.4  | 75.6  | 73.2  | 68.0  | 63.3  | (31) |
|                                                                        |             | MCDB        | 64.9  | 70.3  | 73.9  | 79.1  | 82.2  | 88.4  | 91.0  | 89.1  | 85.1  | 79.3  | 72.7  | 69.9  | (32) |
|                                                                        | 10%         | WB          | 55.4  | 61.9  | 65.1  | 69.3  | 73.4  | 75.9  | 76.3  | 75.5  | 74.8  | 71.6  | 65.7  | 59.6  | (33) |
|                                                                        |             | MCDB        | 63.0  | 67.9  | 71.8  | 76.5  | 81.2  | 86.2  | 88.8  | 87.2  | 83.4  | 78.4  | 70.5  | 64.9  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 22.5  | 21.4  | 22.3  | 22.0  | 20.1  | 20.7  | 22.3  | 22.3  | 20.3  | 21.9  | 21.6  | 20.6  | (35) |
|                                                                        |             | MCDBR       | 29.2  | 25.8  | 24.5  | 25.5  | 23.9  | 23.9  | 25.0  | 24.6  | 24.6  | 23.8  | 23.3  | 25.4  | (36) |
|                                                                        | 5% WB       | MCWBR       | 16.6  | 15.6  | 12.8  | 12.1  | 9.5   | 6.4   | 6.0   | 6.2   | 7.6   | 10.2  | 12.6  | 16.3  | (37) |
|                                                                        |             | MCWBR       | 23.8  | 21.2  | 19.8  | 20.4  | 18.8  | 19.4  | 22.2  | 21.6  | 19.2  | 19.2  | 18.5  | 21.2  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.310 | 0.319 | 0.346 | 0.386 | 0.420 | 0.415 | 0.426 | 0.418 | 0.389 | 0.362 | 0.334 | 0.313 | (40) |
|                                                                        |             | taud        | 2.523 | 2.501 | 2.439 | 2.350 | 2.309 | 2.365 | 2.375 | 2.376 | 2.405 | 2.469 | 2.483 | 2.517 | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 293   | 300   | 298   | 287   | 277   | 277   | 273   | 274   | 279   | 281   | 281   | 285   | (42) |
|                                                                        |             | Edh at Noon | 27    | 30    | 35    | 40    | 42    | 39    | 39    | 38    | 36    | 31    | 28    | 26    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 976   | 1157  | 1480  | 1780  | 1872  | 2137  | 2150  | 2001  | 1662  | 1340  | 1024  | 865   | (44) |
|                                                                        | RadStd      | RadStd      | 104   | 180   | 131   | 164   | 183   | 164   | 168   | 121   | 146   | 199   | 128   | 116   | (45) |

## Historical Trends

|                              |     | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|-----|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |     |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)                          | (b) | (c)   | (d)     | (e)    | (f)   | (g)     | (h)   | (i) | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (43 neighbors) |     |       | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | +135  |

Nomenclature: See separate page